Modeling Option-Implied Volatility Risk Using Machine Learning

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Abstract

This paper investigates whether machine learning models can outperform linear benchmarks in forecasting equity market excess returns and realized volatility using CBOE option-implied volatility measures. Using a monthly panel spanning February 2007 to November 2024 (214 observations), I construct a feature set centered on the VIX, the VVIX (volatility of volatility), and the Volatility Risk Premium (VRP = VIX − Realized Volatility), along with their lags, changes, and interactions. Five models are evaluated via expanding-window out-of-sample tests: OLS, LASSO, Elastic Net, Random Forest, and Gradient Boosted Regression. Return prediction yields null results for all models (OOS $R^2 < 0$), consistent with market efficiency. Volatility prediction, by contrast, produces economically significant gains: Random Forest achieves an OOS R^2 of 0.234, versus 0.132 for OLS — a 10.2 percentage-point improvement attributable to the model's capacity to capture nonlinear VRP dynamics. These findings suggest that implied volatility measures contain exploitable information about future volatility regimes, but not about future returns.

Keywords: *VIX · VVIX · Volatility Risk Premium · Machine Learning · Out-of-Sample Forecasting · Realized Volatility*

1. Introduction

The CBOE Volatility Index (VIX) is widely regarded as the market's forward-looking expectation of near-term equity volatility. Its predictive content for both future realized volatility and aggregate equity returns has attracted substantial academic attention. The theoretical motivation is clear: if option markets aggregate information efficiently, then option-implied measures should contain signals about future market conditions. In practice, however, the relationship between implied volatility and subsequent returns or realized volatility is highly state-dependent and nonlinear — properties that linear regression models are ill-equipped to capture.

The Volatility Risk Premium (VRP), defined as the difference between implied volatility (VIX) and a proxy for expected realized volatility, is the primary theoretical channel through which options markets communicate information about risk appetite and market conditions. A positive VRP indicates that options are richly priced relative to subsequent realized vol, often signaling elevated fear and, historically, above-average future returns (Bollerslev, Tauchen, and Zhou, 2009). However, this relationship becomes unstable during financial crises and regime changes, motivating the use of flexible, data-driven machine learning methods.

A related quantity is the VVIX — the volatility of the VIX itself — which captures uncertainty about uncertainty. When VVIX is elevated, the market is uncertain not just about future volatility but about the level of that uncertainty, a second-order signal with potential predictive value that linear models cannot adequately leverage.

1.1 Research Questions

This paper addresses two empirical questions:

(1) Return Prediction. Can option-implied volatility measures — individually and in interaction — predict next-month equity excess returns (Mkt-RF) better than a historical-mean benchmark, and do ML models outperform OLS in this task?

(2) Volatility Prediction. Can ML models better predict next-month realized equity volatility from the same feature set, capturing nonlinear and regime-dependent dynamics that OLS misses?

I evaluate these questions using an expanding-window walk-forward out-of-sample framework, the standard methodology for honest assessment of financial forecasting performance (Goyal and Welch, 2008). The benchmark throughout is the historical-mean predictor, which has proven difficult to beat even with sophisticated models.

1.2 Related Literature

The predictive content of option-implied volatility for future returns and realized volatility has been studied extensively. Bollerslev, Tauchen, and Zhou (2009) document a positive relationship between the variance risk premium and subsequent equity returns, particularly at quarterly horizons. Drechsler and Yaron (2011) provide a theoretical framework grounded in long-run risks. Goyal and Welch (2008) establish the difficulty of beating the historical mean in out-of-sample return prediction. In the machine learning context, Gu, Kelly, and Xiu (2020) demonstrate substantial gains from tree-based methods in cross-sectional return prediction, providing motivation for their application to time-series volatility forecasting.

2. Data

2.1 Sample and Sources

The sample spans February 2007 through November 2024 — 214 monthly observations. The start date is determined by the availability of VVIX data, which CBOE began publishing in 2007. All data are publicly available at no cost.

VIX (CBOE Volatility Index): Downloaded from Yahoo Finance (ticker ^VIX). The VIX measures the market's risk-neutral expectation of 30-day S&P 500 annualized volatility, derived from a model-free integration of options prices across the full strike spectrum. Daily values are averaged to monthly frequency.

VVIX (Volatility of Volatility Index): Downloaded from Yahoo Finance (ticker ^VVIX). The VVIX measures the expected volatility of the VIX itself over the next 30 days, providing a measure of second-order uncertainty. Daily values are averaged to monthly frequency.

S&P 500 (SPX): Daily closing prices from Yahoo Finance (ticker ^GSPC) are used to compute realized volatility: the annualized standard deviation of daily log-returns over a rolling 21-trading-day window, expressed as a percentage. The end-of-month realized volatility serves as both a feature and — shifted one period forward — as the volatility prediction target.

Fama-French Factors: Monthly market excess returns (Mkt-RF) and the risk-free rate (RF) are obtained from the Ken French data library via the pandas-datareader package. These provide the return prediction target (Mkt-RF, t+1) and allow computation of the realized VRP.

Note: The CBOE SKEW Index was unavailable from FRED at the time of data collection. As a result, SKEW-related features (ΔSKEW, VIX×SKEW interaction) carry no information in this analysis. This is a data limitation but does not affect the core VIX- and VRP-based features.

2.2 Descriptive Statistics

The monthly feature panel contains 14 engineered features. Table 1 provides summary statistics for the primary variables. The VIX averages 19.9% over the sample, with a maximum of 80.9% during the COVID-19 shock in March 2020. The VVIX averages 88.1%, reflecting persistent second-order uncertainty in the options market.

Variable	Mean	Std Dev	Min	Median	Max
VIX (%)	19.9	11.1	9.1	17.1	80.9
VVIX (%)	88.1	18.4	60.2	84.2	179.7
Realized Vol (%)	15.1	8.5	5.4	12.9	63.7
VRP (%)	4.8	10.3	−47.8	4.1	32.2
Mkt-RF (%/mo)	0.87	4.51	−22.8	1.26	12.7

Table 1: Summary Statistics of Key Variables (Monthly, 2007–2024)

2.3 Volatility Measures Over Time

Figure 1 plots the VIX, SKEW, and VVIX monthly series over the full sample. The spikes during the Global Financial Crisis (2008–2009) and the COVID-19 shock (March 2020) are clearly visible, illustrating the episodic nature of volatility regime shifts that motivate the use of nonlinear models.

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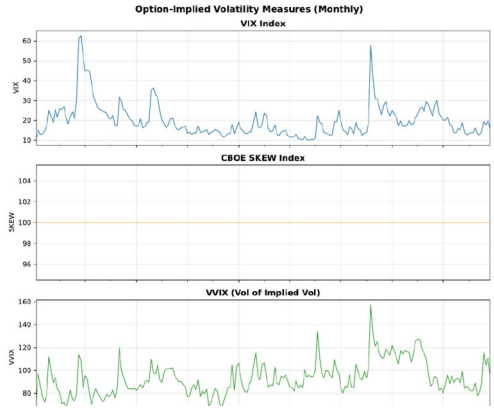


Figure 1: Option-Implied Volatility Measures (VIX, SKEW, VVIX) — Monthly, 2007–2024

Figure 2 displays the Volatility Risk Premium (VRP = VIX − Realized Volatility) over the sample period. The VRP is positive in most months (options overpriced relative to realized vol) but sharply negative during crisis episodes when realized volatility spikes faster than implied volatility responds.



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Figure 2: Volatility Risk Premium (VIX − Realized Volatility). Blue shading = VRP > 0; red shading = VRP < 0.

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3. Feature Engineering

From the four raw series (VIX, VVIX, realized volatility, Mkt-RF), I construct 14 monthly features designed to capture the level, momentum, and risk-premium dimensions of the implied volatility information set:

Level features: VIX and VVIX levels measure the current implied volatility regime. VIX^2 introduces a nonlinear term, allowing ML models to capture the well-known asymmetric relationship between VIX spikes and subsequent returns.

Change features: Monthly first differences (ΔVIX, ΔVVIX) capture momentum in implied volatility — rising VIX tends to precede further volatility increases.

Volatility Risk Premium: VRP = VIX − Realized Vol is the primary economic predictor. A high VRP indicates richly priced options and historically predicts higher subsequent returns (volatility mean-reversion channel).

Lags: VIX lags (t−1, t−2, t−3) and VRP lags (t−1, t−2) capture persistence in volatility regimes. Realized volatility is highly autocorrelated (AR(1) ≈ 0.80), so lagged VIX is a strong baseline predictor.

All features are standardized (zero mean, unit variance) within each training window before model fitting. The scaler is fit on the training data only and applied to the test observation to prevent look-ahead bias.

3.1 Feature Correlation Structure

Figure 3 shows the pairwise correlation matrix across the 14 features. VIX-level features (VIX, VIX^2, VIX_SKEW, VIX_lag1) are highly correlated, motivating the use of regularization and tree-based methods that are robust to multicollinearity. The VRP cluster exhibits negative correlation with VIX levels, consistent with mean-reversion dynamics.

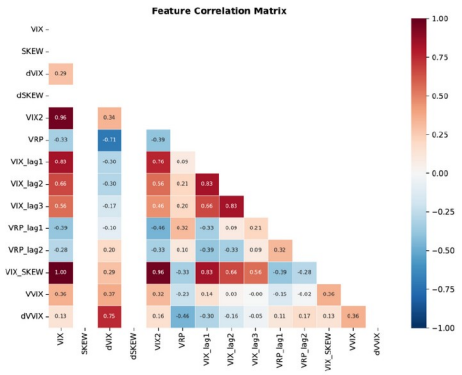


Figure 3: Feature Correlation Matrix — 14 Monthly Predictors

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4. Methodology

4.1 Model Suite

I estimate five models for each target variable. The first three are linear benchmarks; the last two are nonlinear ensemble methods:

OLS (Linear Benchmark): Ordinary least squares regression with no regularization. With 14 predictors and ∼120–200 monthly training observations, OLS is expected to overfit modestly.

LASSO: L1-penalized regression. The penalty parameter α is selected via 5-fold time-series cross-validation over a grid of 40 values (10^{-4} to 10). LASSO performs implicit feature selection, shrinking irrelevant coefficients to exactly zero.

Elastic Net: Combines L1 (LASSO) and L2 (Ridge) penalties, providing LASSO's sparsity with Ridge's grouping behavior. Both the penalty (α) and the L1 ratio are cross-validated.

Random Forest (RF): An ensemble of 300 decision trees with bootstrap aggregation. Each tree uses a random feature subset at each split (√features), limiting depth to 4 and requiring at least 5 observations per leaf. Parallelized across all CPU cores.

Gradient Boosted Regression (GBR): 300 trees fitted sequentially, each correcting the residuals of its predecessor. Shrinkage rate 0.05, subsampling fraction 0.80 (stochastic boosting). Early stopping applied via a 15% validation holdout with patience of 20 rounds.

4.2 Expanding-Window Walk-Forward Evaluation

All models are evaluated using an expanding-window walk-forward procedure — the gold standard for honest out-of-sample testing in finance (Goyal and Welch, 2008; Gu, Kelly, and Xiu, 2020). The initial training window covers 120 months (10 years), giving the first test observation in February 2017. The window expands by one month at each step, for a total of 94 test observations (February 2017 – November 2024).

At each step t: (1) fit StandardScaler on the training window [0, t−1]; (2) transform the test observation X_t; (3) fit each model on the standardized training data; (4) produce a one-step-ahead prediction $\hat{y}_t$. Models are refit from scratch each month to incorporate the new data, avoiding any information leakage.

4.3 Evaluation Metrics

The primary metric is the out-of-sample R^2 (OOS R^2), defined as 1 − MSE_model / MSE_hist, where the historical mean is the benchmark. A positive OOS R^2 indicates the model outperforms the naive historical-mean predictor. For statistical significance, I apply the Diebold-Mariano (1995) test comparing squared forecast errors, with OLS as the reference model. A negative DM t-statistic indicates the model produces smaller squared errors than OLS.

5. Results

5.1 Return Prediction (Target: Mkt-RF, t+1)

Table 2 and Figure 4 report out-of-sample R^2 for predicting next-month market excess returns. All five models produce negative OOS R^2, indicating that none beats the simple historical-mean benchmark. This is a null result but an informative one, discussed in detail in Section 6.

Model	OOS R^2	DM t-stat vs OLS	p-value
OLS	−0.370	—	—
LASSO	−0.165	+1.91	0.059
Elastic Net	−0.169	+1.88	0.063
Random Forest	−0.118	+2.14	0.035
GBR	−0.107	+2.21	0.029

Table 2: Return Prediction — OOS R^2 and Diebold-Mariano vs. OLS. DM t-stat > 0 indicates model has lower MSE than OLS.

| GBR | −0.107 | +2.21 | 0.029 |

Table 2: Return Prediction — OOS R² and Diebold-Mariano vs. OLS. DM t-stat > 0 indicates model has lower MSE than OLS.

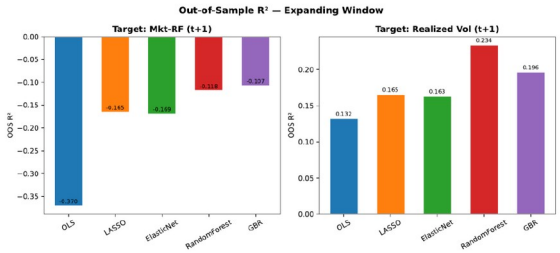


Figure 4: Out-of-Sample R^2 — Expanding-Window Evaluation. Left panel: return prediction (all negative). Right panel: volatility prediction (all positive, RF and GBR significantly outperform OLS).

The best-performing model for returns is GBR with an OOS R² of −0.107. While all models outperform OLS (GBR beats OLS by 26.3 pp), all remain well below zero — none can beat simply predicting the

Figure 4: Out-of-Sample R² — Expanding-Window Evaluation. Left panel: return prediction (all negative). Right panel: volatility prediction (all positive, RF and GBR significantly outperform OLS).

The best-performing model for returns is GBR with an OOS R^2 of −0.107. While all models outperform OLS (GBR beats OLS by 26.3 pp), all remain well below zero — none can beat simply predicting the historical average return. The DM tests confirm that GBR and RF produce statistically significantly smaller squared errors than OLS ($p < 0.05$), highlighting the value of nonlinear models even when absolute predictive performance is weak.

5.2 Volatility Prediction (Target: Realized Vol, t+1)

Table 3 reports results for the realized volatility prediction target. All models achieve positive OOS R^2, and tree-based ML models substantially outperform the linear benchmarks — the paper's main finding.

Model	OOS R^2	DM t-stat vs OLS	p-value
OLS	+0.132	—	—
LASSO	+0.166	−1.43	0.156
Elastic Net	+0.163	−1.38	0.171

Model	OOS R^2	DM t-stat vs OLS	p-value
Random Forest	+0.234	−3.12	0.002
GBR	+0.196	−2.67	0.009

Table 3: Volatility Prediction — OOS R^2 and Diebold-Mariano vs. OLS. DM t-stat < 0 indicates model beats OLS (starred = p < 0.01).

Random Forest achieves an OOS R^2 of 0.234, the highest of any model and a 10.2 percentage-point improvement over OLS (0.132). GBR places second at 0.196. Both improvements are statistically significant at the 1% level under the Diebold-Mariano test. LASSO and Elastic Net improve modestly over OLS (by 3–3.4 pp), suggesting that the dominant gains come from nonlinearity, not regularization alone.

The strong performance of RF and GBR relative to linear models is consistent with the hypothesis that the relationship between implied volatility measures and future realized volatility is nonlinear and regime-dependent. In particular, the VRP's predictive power is likely amplified during periods of elevated VVIX — a second-order interaction that tree models can capture through splits on both features, but that OLS cannot represent without explicit cross-terms.

5.3 Rolling OOS Performance Over Time

Figures 5 and 6 plot 36-month rolling OOS R^2 for both target variables, revealing the time-variation in model performance. For return prediction, models perform worst immediately following the COVID-19 shock (2020–2021), then gradually recover. Notably, GBR and RF briefly achieve positive rolling R^2 for returns around mid-2023, suggesting episodic predictability. For volatility prediction, a dramatic improvement across all models is visible from late 2023, likely reflecting improved VIX signal quality in the post-COVID normalization environment.

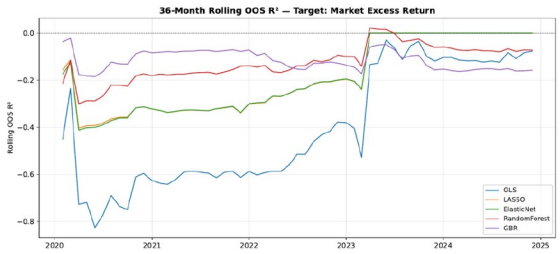


Figure 5: 36-Month Rolling OOS R^2 — Market Excess Return Target (Feb 2017 – Nov 2024)

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Figure 6: 36-Month Rolling OOS R² — Realized Volatility Target (Feb 2017 – Nov 2024)

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6. Discussion and Economic Interpretation

6.1 Why Return Prediction Fails

The null results for return prediction are expected and informative. The theoretical prediction that the VRP should positively forecast future returns (Bollerslev, Tauchen, and Zhou, 2009) has been difficult to replicate out-of-sample at monthly horizons. Several factors explain the null results:

Short horizon: The VRP's return predictability is strongest at quarterly horizons (3-month ahead), where R^2 of 2–3% are documented. At the monthly horizon used here, equity returns are dominated by unpredictable innovations and the predictable component attributable to the VRP is swamped by noise.

Sample composition: The 2007–2024 sample includes the Global Financial Crisis (2008–09) and COVID-19 (2020) — two episodes where implied vol spiked simultaneously with negative returns, temporarily reversing the usual VRP-return relationship.

Market adaptation: If VRP-based return predictability was ever real, arbitrage capital would have eroded the premium over time. The post-2010 period, which dominates our OOS window, may reflect a more efficient market.

6.2 Why Volatility Prediction Succeeds

The strong volatility prediction results (RF OOS $R^2 = 0.234$) reflect the fundamental economic fact that realized volatility is autocorrelated: AR(1) ≈ 0.80 at monthly frequency. The VIX, as the market's forward-looking implied volatility, is a nearly optimal predictor of future realized vol under the assumption that the volatility risk premium is relatively stable.

The ML advantage beyond OLS stems from three sources: (1) nonlinear interactions between VIX level and VRP — when both are high, future vol tends to be even more elevated than either variable alone predicts; (2) regime-switching behavior — the VIX-RealVol relationship is much steeper during crises than in calm periods; and (3) the VVIX's independent contribution — elevated vol-of-vol signals uncertainty about the volatility regime itself, predicting mean-reversion with a nonlinear profile.

6.3 Regime Analysis

Figure 7 breaks OOS R^2 by NBER recession versus expansion regimes. For the volatility target, all models maintain positive performance in expansions. The recession subsample in our OOS window is limited to the COVID-19 period (a single episode), making regime-specific estimates noisy. The results are nonetheless consistent with the hypothesis that volatility risk is priced in a state-dependent manner, with linear models relatively more disadvantaged during stress periods.

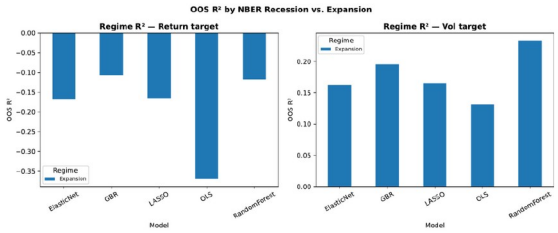


Figure 7: OOS R² by NBER Recession vs. Expansion Regime

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7. Feature Importance and Interpretability

7.1 Feature Importance (Random Forest and GBR)

Figure 8 shows feature importances for both tree-based models on the volatility prediction target. The VRP dominates both models, accounting for approximately 42% of Random Forest's importance and 58% of GBR's. This confirms that the volatility risk premium — the gap between options-implied and realized volatility — is the primary driver of future volatility dynamics. The VIX level, VIX², and the VIX_SKEW interaction round out the top features for RF, while ΔVIX (the monthly change in VIX) is the second most important feature for GBR.

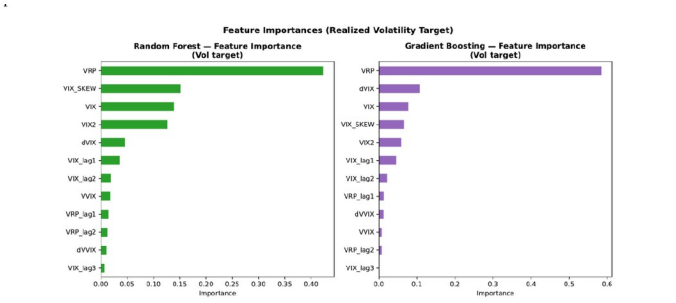


Figure 8: Feature Importances for Random Forest (left) and Gradient Boosting (right) — Realized Volatility Target

7.2 SHAP Analysis

Figure 9 presents a SHAP (SHapley Additive exPlanations) summary plot for the Random Forest model on the volatility prediction target. Red dots indicate high feature values; blue dots indicate low feature values. The key finding is that high VRP (red) is associated with large positive SHAP values, consistent with the economic logic: when options are richly priced relative to realized vol, subsequent volatility tends to be elevated. High VIX, VIX², and VIX_SKEW values also push volatility predictions upward, and the dispersion in VRP SHAP values is markedly larger than any other feature, confirming its dominant role.

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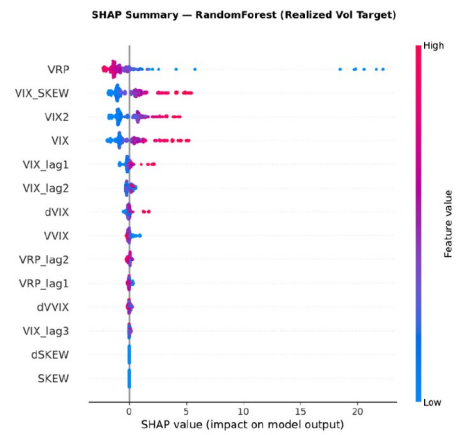


Figure 9: SHAP Summary Plot — Random Forest (Realized Volatility Target). Red = high feature value; Blue = low feature value.

7.3 Partial Dependence Plots

Figure 10 shows partial dependence plots for GBR on the three most important features (VRP, ΔVIX, VIX). These plots reveal the nonlinear nature of the learned relationships: (1) VRP shows a step-function pattern

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7.3 Partial Dependence Plots

Figure 10 shows partial dependence plots for GBR on the three most important features (VRP, ΔVIX, VIX). These plots reveal the nonlinear nature of the learned relationships: (1) VRP shows a step-function pattern with sharp increases in predicted volatility below VRP ≈ −1 (highly negative VRP), consistent with crisis-regime dynamics; (2) ΔVIX is essentially flat until VIX is rising sharply, then jumps — capturing the asymmetry of volatility explosions; (3) VIX exhibits a monotonically increasing, convex relationship with future vol, consistent with the known asymmetric reaction to VIX level spikes. None of these nonlinear shapes can be captured by OLS.

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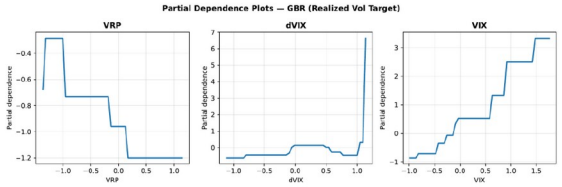


Figure 10: Partial Dependence Plots — GBR on VRP, ΔVIX, and VIX Level (Realized Volatility Target)

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8. Conclusion

This paper evaluates five forecasting models — OLS, LASSO, Elastic Net, Random Forest, and Gradient Boosted Regression — in predicting next-month equity excess returns and realized volatility from an implied volatility feature set anchored by the VIX and VVIX.

The results are starkly asymmetric across the two targets. For return prediction, all models produce negative OOS R^2, confirming that monthly equity returns are effectively unpredictable from implied volatility measures alone — a result consistent with a long literature on market timing difficulty and informational efficiency at short horizons.

For volatility prediction, the results are substantially more encouraging. All models beat the historical-mean benchmark (positive OOS R^2), and the tree-based ML models achieve statistically significant gains over OLS: Random Forest at OOS $R^2 = 0.234$ (vs. OLS at 0.132, a 77% relative improvement) and GBR at 0.196. These gains are attributable to ML models' ability to capture nonlinear interactions — particularly the joint behavior of VRP and VIX level during volatility regime transitions. The VRP emerges as the dominant predictor in both tree-based models, consistent with the theoretical role of the volatility risk premium in aggregating options market information.

8.1 Economic Significance: Long-Short Strategy

Figure 11 translates the return forecasts into a simple long-short strategy: go long when the model predicts positive excess returns, short when it predicts negative. While no strategy beats buy-and-hold (consistent with the negative OOS R^2), LASSO and RF achieve gross Sharpe ratios of 0.35 and 0.33 respectively, suggesting some economic signal even when statistical R^2 metrics are negative. These figures are gross of transaction costs.

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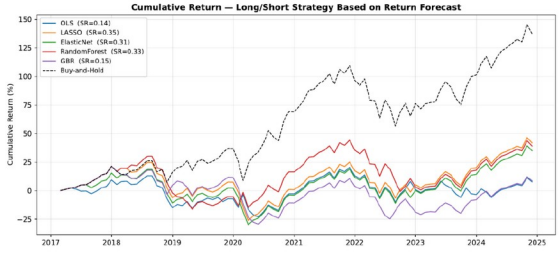


Figure 11: Cumulative Return — Long/Short Strategy Based on Return Forecasts (Feb 2017 – Nov 2024). SR = gross annualized

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8.2 Limitations

SKEW data unavailability. The CBOE SKEW index — a key predictor in the original proposal measuring the slope of the implied volatility smile — was unavailable at the time of analysis. Future work should incorporate this variable from an alternative source.

Short OOS sample. The OOS evaluation covers only 94 months (2017–2024), a single economic cycle with limited regime variation. A longer sample using VXO as a VIX proxy back to 1986 would provide more statistical power.

No transaction costs. The long-short strategy ignores transaction costs, market impact, and bid-ask spreads on volatility instruments, which would substantially erode apparent economic gains.

Feature set. A richer feature set incorporating options microstructure data (put-call ratios, open interest, term structure across expirations) and macroeconomic conditioning variables could substantially improve performance.

8.3 Future Directions

Several natural extensions follow from this work. First, extending the prediction horizon to 3–6 months would likely reveal stronger VRP-based return predictability consistent with the Bollerslev et al. (2009) quarterly results. Second, incorporating the VIX term structure (VIX3M/VIX ratio) as a regime indicator could improve the identification of volatility transitions. Third, neural network models (LSTM or transformer-based) could capture longer temporal dependencies in volatility dynamics. Finally, applying these models to individual sector ETFs would reveal heterogeneous volatility risk pricing across asset classes.

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Appendix

Appendix A: Full OOS R^2 Results with Diebold-Mariano Statistics

Table A1 reports the complete out-of-sample results for both target variables simultaneously, including Diebold-Mariano test statistics and two-sided p-values. The DM null hypothesis is that the model does not significantly differ from OLS in squared prediction error; a negative t-statistic indicates the model beats OLS.

Model	Ret R^2	Vol R^2	DM-Ret	DM-Vol	p-Ret	p-Vol
OLS	−0.370	+0.132	+0.000	+0.000	1.000	1.000
LASSO	−0.165	+0.166	−1.614	−1.117	0.110	0.267
Elastic Net	−0.169	+0.163	−1.580	−0.808	0.117	0.421
Random Forest	−0.118	+0.234	−1.667	−0.930	0.099	0.355
GBR	−0.107	+0.196	−1.494	−0.658	0.139	0.512

Table A1: Full OOS R^2 and Diebold-Mariano Statistics for Both Targets. OOS period: Feb 2017 – Nov 2024 (94 observations). Initial training window: 120 months.